

Question 1 – Bike Sharing Dataset Exploratory Analysis

Q1 Summary Report

Introduction

This report summarises our findings from exploratory data analysis into a bike sharing dataset. This dataset spans bike rental counts, as well as other impactful factors across 2011 and 2012. From our understanding and searching, the dataset originates from the Capital Bikeshare system in Washington, D.C. and can be accessed on Kaggle. As instructed by the assignment, we chose to focus on three objectives: identifying key relationships, detecting outliers, and identifying anomalies or irregular structure.

Key Relationships

We identified various different relationships during our analysis. These are listed and discussed below:

Temperate-Usage relationship. Temperature was the most impactful driver in bike rental counts across our dataset. The correlation between normalised temperature and total rentals is approximately $r = 0.63$. We also found the relationship to be non linear, the number of rentals increases steadily up until a normalised temperature of around 0.7, before declining. This indicates that an increase in heat/temperature is a driving factor, until the temperature feels too extreme, in which case users would prefer other transportation methods to bike rentals.

Humidity-Wind-Usage relationship. Humidity and wind are negatively correlated with usage, though weaker than temperature. Humidity showed an increase in usage, up until a normalised humidity of approximately $r = 0.75$, where a steep sharp decline would occur. This would make sense, as a mild increase in humidity could correspond with warmer weather, while very high humidity would correspond with heavy precipitation or snowfall. Wind speed saw a near linear negative relationship, where an increase in wind speed leads to a decrease in usage. The categorical weather indicator also seems to support these relationships, as we increase from 1 to 3, we see less usage. Interestingly, category 4, representing the most severe conditions, is not present in the dataset. This seems to be a data error.

Weekend-Weekday user pattern. Across weekdays and weekends, the total number of rentals are broadly similar. Decomposing this further into the two different type of rentals, casual and registered users, reveals a substantial pattern. Casual users peak on non-working days (weekends), aligned with recreational use, while registered users peak on working days (weekdays), indicating the users are commuting.

Systematic year-on-year growth. 2012 shows consistently higher usage rates than 2011. Plotting a month by month comparison, we can see that both years hold the same consistent seasonal shape, but 2012 reports an increase in users month by month, indicating a systematic shift in growth, rather than a seasonal pattern.

Outlier Detection

We tried to perform standard statistical outlier tests, that of Modified Z-Scores and IQR rules. However, both of these methods failed to flag any outliers, as the distribution of the rentals was very wide. As there were still obvious outliers, we choose to select outliers based on a domain based simple method instead, taking 15% of the median value as the cut off point. This resulted in seven outliers, all on the lower tail. Investigating these further, we noted that most of these corresponded to historical weather events in Washington, D.C. One of the outliers had a 0 humidity recorded, but this seems to be a data error, and the outlier would've been flagged regardless. These events are shown in the table below.

Date	cnt	Likely Cause
2011-01-26	506	Major snowstorm ("Carmageddon")
2011-01-27	431	Aftermath of Carmageddon snowstorm
2011-03-06	605	Near-maximum humidity (0.95) — heavy rainfall
2011-03-10	623	Severe thunderstorms
2011-10-29	627	Halloween Nor'easter ("Snowtober") — early-season snowstorm
2012-10-29	22	Hurricane Sandy — federal government closed, 55–65 mph winds
2012-12-26	441	Day after Christmas — holiday period low usage

Anomalies, Structural Irregularities and Data Issues

Several anomalies were found that required further investigation:

Near perfect collinearity between temp and atemp. Temperature and apparent temperature are nearly perfectly collinear with an approximate $r = 0.99$. While this makes sense, as apparent temperature is derived from temperature, if further analysis or investigation was to take place, you would have to remove one of these factors in order to avoid introducing multicollinearity. As apparent temperature is derived from temperature, it makes the most sense to retain temperature.

Humidity Data Point 2011-03-10. For the 4th outlier found, we reported a humidity of 0. This is almost surely a data issue, as searching online seems to suggest there were storms in the area. Further investigation showed that there was only one data point with a humidity of 0, so there are no structural irregularities, but this is an anomaly that may be considered if modelling was to take place as it could significantly impact humidity predictors.

"weathersit" – weather category 4. Weather category 4, which represents the most severe conditions, does not appear in the dataset. Even days that have documented severe weather events, such as that within our outlier sets, are documented as category 3. This suggests that there is a structural irregularity, and that extreme weather events are categorised incorrectly.

Conclusion

To conclude, our exploratory data analysis provided enough evidence to support the production of a model based on bike rental usage, and weather driven factors. Primarily, weather usage is driven by temperature, but also by secondary effects from humidity, wind speed and weather conditions. Day-by-day and seasonal patterns were identified, as well as systematic yearly growth. Seven outliers were identified, a quality data issue as well as a potential structural data mislabelling issue.

Question 2 – The tidyverse approach

The tidyverse is a collection of R packages that share a common design philosophy, and enable users to standardise the methods and function used for exploring, cleaning and manipulating data. The packages are based on the tidy data philosophy, where each variable forms a column and each row an observation. Its ease of use, wide industry appeal and standardised formatting give it a variety of strengths.

The main strength is the readability provided from using tidyverse packages and methods. In comparison to R's often verbose and complicated syntax, tidyverse provides consistent verb-based grammar, and regimented syntax through its pipe operator. For example, in Q1 of the assignment, we prep the dataset by converting columns and relabelling factors. We do this using `day_data %>% mutate(...) %>% filter(...)`, which intuitively allows you to understand the methods being applied. Compared to R's base language, which would've required many nested functions, repeated calls, or temporary variables – severely limiting readability. Cleaner, easily readable code is safer, reducing errors through clarity and ensuring checks during EDA can be performed quickly and efficiently.

This system carries onto other methods, not just modifying data. When plotting using ggplot2 functions, we have a standardised, reproducible and modifiable syntax. The same skeleton format to plot a bar chart can easily be transformed to plot scatterplots with smoothers. Each different plot requires a small addition. This allows us to easily replicate style across plots, and allows incredibly simple bug-fixes and modifications through the hyper-compartmentalised nature of the syntax.

However, the system is not without its faults. As encountered in Q1, the interdependencies across packages can lead to significant setbacks if not managed properly. Each package has its own update cycle, and modifications to packages can often cause breakage and failures, leading to long troubleshooting sessions. Its important that R packages are managed correctly before hand, particular if you need to use tidyverse. In Q1, I was unable to combine plots without installing the patchwork library. This led to error messages, and took some troubleshooting to solve because of some dependency conflicts, while using base R would've required no additional work.

Similarly, tidyverse's biggest advantage can also become a weakness. Too much abstraction and generality in its syntax can lead to non-standard evaluation. This leads to column names being passed in irregular formats, such as unquoted. While we did not come across this in Q1, if a function needed to accept arbitrary columns, excess syntax would need to be used, complicating the methodology, and negating tidyverse's benefits. Another passing note, the abstraction carries overhead to produce the generality, and can lead to performance issues, in particular with large datasets.

To end, if used correctly, tidyverse is a powerful collection of packages that can produce readable and precise code, but if its disadvantages are not mitigated, it can lead to more confusion and hassle for both the developer and reader.